

A Quantitative Analysis of the Dissipation Inherent in Semi-Lagrangian Advection

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ABSTRACT

Semi-Lagrangian advection schemes are known to be dissipative because of the interpolation required to estimate the values of the flow fields at each parcel's departure point. In this study, the amplification factors of first, second, third, and fourth-order Lagrangian interpolation schemes are used to calculate the dissipative decay time scale and the resulting effective eddy viscosity as functions of wavelength and residual Courant number. The dissipation inherent in the semi-Lagrangian advection can then be compared to more traditional forms of dissipation, such as Laplacian or biharmonic eddy viscosity. The correspondence between semi-Lagrangian advection and more traditional Eulerian techniques is emphasized. The dependence of the dissipation on the time step and grid spacing is also discussed, with a view to selecting the discretionary parameters to meet conservation criteria.

1. Introduction

In the search for more efficient numerical techniques for the integration of the atmospheric equations of motion, a number of authors have recently been experimenting with the so-called semi-Lagrangian advective scheme. This scheme works by back-tracking along particle trajectories and interpolating on the flow fields to update the values at the fixed grid points. It differs from a pure Lagrangian scheme in that the parcels followed are different at each time step. When the grid points used in the interpolation are the nearest neighbors to the parcel's departure point (rather than its arrival point), then the scheme is called "multiply-upstream." Advection with a multiply-upstream scheme is unconditionally stable.

There are many options in the implementation of semi-Lagrangian advection. Several authors have used a split scheme (Bates and McDonald 1982; Bates 1984), in which the equations are divided into pure advection and linear relaxation, and integrated independently. Then the time step for the advective portion can be based on accuracy considerations, rather than stability criterion. Alternatively, the equations may be integrated in a single step, with the nonadvective terms being moved to the right-hand-side and evaluated along the trajectory (Robert et al 1985; McDonald 1986). For simplicity, the analysis here is for a pure advective scheme.

McDonald (1984, 1987) has discussed the accuracy of semi-Lagrangian advection using Lagrangian interpolation schemes. He and others have noted its dissipation (see Bates 1984; Robert et al 1985; Ritchie 1988), but have not quantified it in terms of more common measures of viscosity. In this study, the dissipation inherent in semi-Lagrangian advection will be quantified in terms of the dissipative decay time scale and the effective eddy viscosity as functions of the horizontal scale and the residual Courant number.

2. Amplification factors

McDonald (1984) has calculated the amplification factors for a particular implementation of the semi-Lagrangian advection of plane waves using Lagrangian interpolation. The amplification factor of the Lagrangian interpolation scheme is separable into multiplicative x and y components, which makes the analysis much simpler. Since we are only concerned with the amplitudes here (not the phase), it is sufficient to deal with only the magnitudes of the amplification factors. Following McDonald, the interpolation points are chosen such that the fractional part of the Courant number lies between 0 and 1 for the cases of bilinear and bicubic interpolation, and between $-1/2$ and $+1/2$ for the cases of biquadratic and biquartic interpolation. Hereafter, the fractional part of the Courant number will be referred to by α , and will be called the "residual Courant number." With the definition: $c = 1 - \cos(k\Delta x)$, the magnitudes of the amplification factors for the four schemes are thus:

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$$\left. \begin{aligned} |\lambda_1|^2 &= 1 - 2\alpha(1 - \alpha)c \\ |\lambda_2|^2 &= 1 - \alpha^2(1 - \alpha^2)c^2 \\ |\lambda_3|^2 &= 1 - \alpha(2 - \alpha)(1 - \alpha^2)c^2 \\ &\quad \times [3 + 2c\alpha(1 - \alpha)]/9 \\ |\lambda_4|^2 &= 1 - \alpha^2(1 - \alpha^2)(4 - \alpha^2)c^3 \\ &\quad \times [4 + c(1 - \alpha^2)]/36 \end{aligned} \right\}, \quad (2.1)$$

where α is the residual Courant number, k the wavenumber, and Δt and Δx are the usual finite difference time and space increments, respectively. The subscript indicates the order of the interpolation.

Needless to say, these are not the only possible choices for the interpolation. Spline interpolations are known to have some properties that are superior to Lagrangian interpolation, and have been used with semi-Lagrangian advective schemes (Purnell 1976; Pudykiewicz and Staniforth 1984). Low-order trigonometric interpolations have also been investigated for semi-Lagrangian advection in storm surge modeling (R. O. Reid, personal communication). Therefore, the details of a particular implementation may differ from the results presented here, but the orders of accuracy should be the same for varying interpolation techniques.

3. Decay time scales and "equivalent" viscosity

The amplification factors shown above all have magnitudes less than or equal to unity, ensuring stability for the schemes. However, the magnitude of the amplification factor is not a very understandable measure of the dissipation in a finite-difference scheme (unless it is significantly smaller than 1!). A much more easily understood measure is the dissipative decay time scale. This is defined as the time required to reduce a signal to e^{-1} times its original amplitude. In terms of the amplification factor, the decay time scale is

$$\tau = -\frac{\Delta t}{\ln|\lambda|}. \quad (3.1)$$

An alternate representation of the magnitude of the dissipation is the coefficient of viscosity or hyperviscosity which produces the same damping. The dependence of the damping on the wavelength will show how the dissipation is more or less scale dependent than the traditional Laplacian eddy viscosity formulation, which has a k^2 spectral dependence.

The dependencies of the decay time scales on α and wavelength for each of the four schemes described in the previous section are shown in Figs. 1–4. Note that the α axis is linear, and that the wavelength axis and the contour interval are logarithmic. Note also that since the wavenumber dependence of the decay time scale appears only as $k\Delta x$, the vertical axis has been nondimensionalized by Δx . The grid interval Δx also

shows up in the residual Courant numbers, so the effects of time and space discretization cannot be entirely separated. This will be discussed in some detail in section 5. The decay time scale is in units of days, for a time increment $\Delta t = 3600$ s. When it is appropriate to use dimensional quantities, the values $\Delta x = 200$ km and $\Delta t = 3600$ s will be used as representative numbers. For this "typical" horizontal resolution the figures cover the two decades from the Nyquist cutoff on the short-wave end, to the global mode 1 wave on the long-wave end. The following sections will discuss the long-wave and short-wave characteristics of the semi-Lagrangian scheme separately.

a. Long waves

For long waves ($k\Delta x \ll 1$), the wavenumber dependent term in the amplification factors $[1 - \cos(k\Delta x)]$ reduces to approximately $\frac{1}{2}(k\Delta x)^2$. Inspection of (2.1) shows that linear interpolation has the same spectral characteristics as Laplacian viscosity and quadratic and cubic interpolation have the same spectral characteristics as biharmonic viscosity.

It may often be the case that the advecting velocity is only large in a portion of the computational domain. For the regions in which the Courant number is small, the amplification factors can be approximated:

$$\left. \begin{aligned} |\lambda_1| &= 1 - \frac{1}{2}\alpha(k\Delta x)^2 \\ |\lambda_2| &= 1 - \frac{1}{8}\alpha^2(k\Delta x)^4 \\ |\lambda_3| &= 1 - \frac{1}{12}\alpha(k\Delta x)^4 \\ |\lambda_4| &= 1 - \frac{1}{36}\alpha^2(k\Delta x)^6 \end{aligned} \right\}. \quad (3.2)$$

This shows clearly that the third-order scheme can have more dissipation than the second-order scheme, and that the spectral dependence of its diffusivity is the same. The third-order scheme does, however, have a better phase representation than the second-order scheme (as pointed out by McDonald 1984).

Purnell (1976) and Pudykiewicz and Staniforth (1984) analyzed a cubic-spline interpolation from which can be obtained an analogous expression for $|\lambda_3|$, which reduces to

$$|\lambda_3| = 1 - \frac{1}{12}\alpha^2(k\Delta x)^4,$$

in this limit. This result shows that the cubic-spline interpolation has slightly less dissipation than the quadratic Lagrangian interpolation used here, but displays the same spectral slope.

For these long waves, it is straightforward to calculate the equivalent eddy viscosity from the amplification

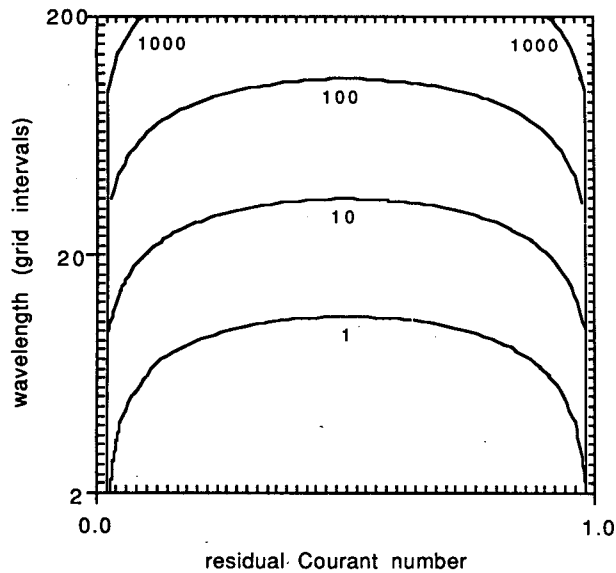


FIG. 1. Contours of the dissipation decay time scale in days as a function of residual Courant number ($U\Delta t/\Delta x$) and wavelength for the case of one-dimensional semi-Lagrangian advection with linear interpolation. Note that the vertical axis and contour interval are logarithmic. The vertical contours near $\alpha = 0$ and $\alpha = 1$ are artifacts of the contouring.

factor. The analytic solution for the amplitude of a plane wave damped by Laplacian friction is

$$\hat{\psi}(k, t) = \hat{\psi}(k, 0)e^{-A_H k^2 t}.$$

The Taylor series expansion (in time) for the amplification factor is

$$|\lambda| = 1 - A_H k^2 \Delta t + O(\Delta t^2).$$

Then, equate the first two terms of the long-wave ex-

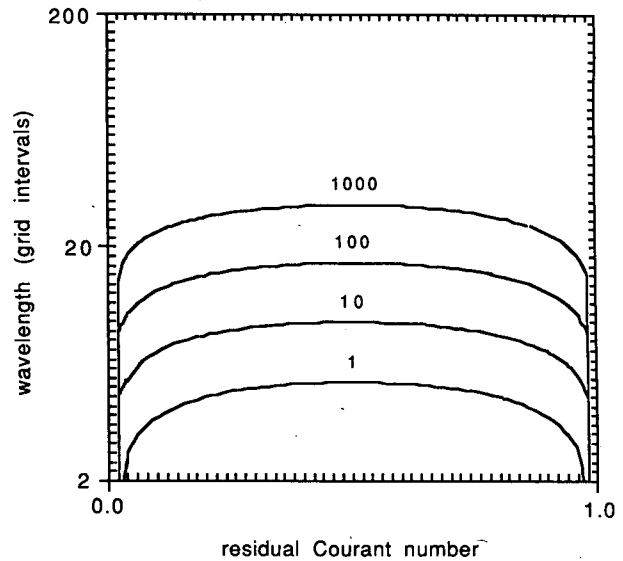


FIG. 3. As in Fig. 1, but for cubic interpolation.

pansion for the semi-Lagrangian step with these two terms and solve for A_H . The results are

$$A_{H1} = \frac{1}{2\Delta t} (\alpha - \alpha^2) \Delta x^2$$

$$A_{H2} = \frac{1}{8\Delta t} (\alpha^2 - \alpha^4) \Delta x^2 (k\Delta x)^2$$

$$A_{H3} = \frac{1}{24\Delta t} (2\alpha - \alpha^2 - 2\alpha^3 + \alpha^4) \Delta x^2 (k\Delta x)^2$$

$$A_{H4} = \frac{1}{144\Delta t} (4\alpha^2 - 5\alpha^4 + \alpha^6) \Delta x^2 (k\Delta x)^4$$

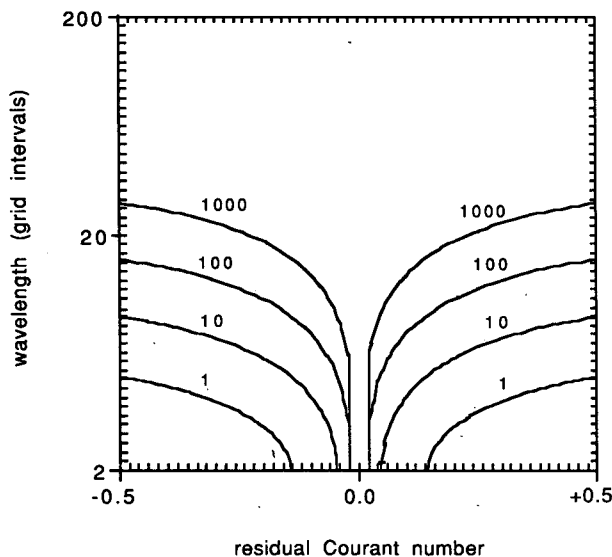


FIG. 2. As in Fig. 1, but for quadratic interpolation.

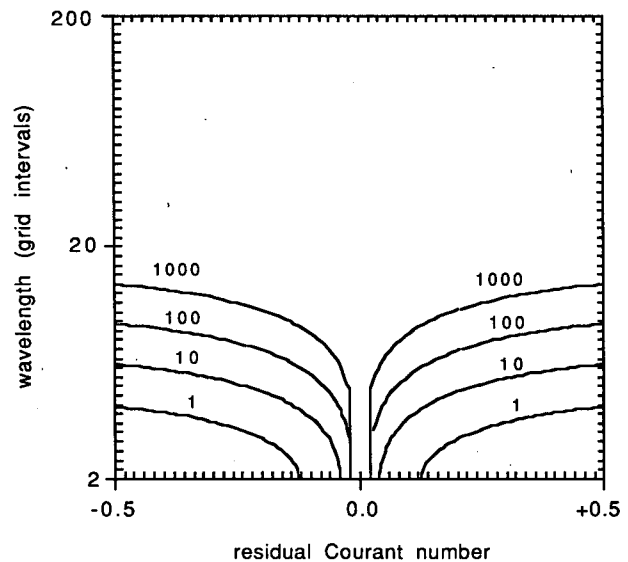


FIG. 4. As in Fig. 1, but for quartic interpolation.

Only the linear case has the k^2 spectral dependence characteristic of Laplacian friction. Since α is in the range 0–1, an upper bound to the equivalent Laplacian dissipation for this case is

$$A_{H1} \leq \frac{1}{2\Delta t} \left(\frac{1}{4} \right) \Delta x^2 = \frac{1}{8} \frac{\Delta x^2}{\Delta t}, \quad (3.3)$$

which is approximately $1.4 \times 10^6 \text{ m}^2 \text{ s}^{-1}$ for the parameters listed at the beginning of this section. For the atmosphere, values of the eddy diffusion coefficient have historically been estimated at less than $10^6 \text{ m}^2 \text{ s}^{-1}$, though it is generally accepted that Laplacian eddy viscosity is a rather poor approximation to the actual turbulent dissipation and diffusion in the atmosphere.

An analogous comparison may be made for biharmonic viscosity. The equivalent biharmonic viscosities are

$$A_{B1} = \frac{1}{2\Delta t} (\alpha - \alpha^2) \Delta x^4 (k\Delta x)^{-2}$$

$$A_{B2} = \frac{1}{8\Delta t} (\alpha^2 - \alpha^4) \Delta x^4$$

$$A_{B3} = \frac{1}{24\Delta t} (2\alpha - \alpha^2 - 2\alpha^3 + \alpha^4) \Delta x^4$$

$$A_{B4} = \frac{1}{144\Delta t} (4\alpha^2 - 5\alpha^4 - \alpha^6) \Delta x^4 (k\Delta x)^2.$$

Noting the ranges available for α , the upper bounds for the quadratic and cubic cases are

$$\begin{aligned} A_{B2} &\leq \frac{1}{8\Delta t} \left(\frac{3}{16} \right) \Delta x^4 = \frac{3}{128} \frac{\Delta x^4}{\Delta t} \\ A_{B3} &\leq \frac{1}{24\Delta t} \left(\frac{9}{16} \right) \Delta x^4 = \frac{3}{128} \frac{\Delta x^4}{\Delta t} \end{aligned} \quad (3.4)$$

which are both approximately $1.0 \times 10^{16} \text{ m}^4 \text{ s}^{-1}$ for the parameters listed at the beginning of this section. It is difficult to define this nonphysical dissipation observationally, but the European Center for Medium-Range Weather Forecasting high-resolution model uses values of about $1.0 \times 10^{15} \text{ m}^4 \text{ s}^{-1}$ for medium range forecasts (see Ritchie 1988).

b. Short waves

For short waves, the term $[1 - \cos(k\Delta x)]$ does not display the same dependence on k as for the long waves. Two cases of special interest are wavelengths of $2\Delta x$ and $4\Delta x$, for which the cosine terms reduce to 2ⁿ and 1, respectively, where n is the order of the interpolation. The dissipation still increases with increasing wavenumber, but it does so at a much slower rate than the appropriate power of k near the shortest waves in the system. Thus, the equivalent eddy viscosity actually decreases for the shortest waves, and the effect gets

worse as the order of the scheme increases. This is unfortunate, as one would usually desire larger dissipation in this wavelength range to damp out the poorly resolved (and inaccurate) short waves.

It is instructive to examine the amplification factors for these special cases, which are simply polynomials in α :

$$L = 2\Delta x$$

$$|\lambda_1|^2 = 1 - 4\alpha + 4\alpha^2$$

$$|\lambda_2|^2 = 1 - 4\alpha^2 + 4\alpha^4$$

$$|\lambda_3|^2 = 1 - \frac{8}{3}\alpha - \frac{20}{9}\alpha^2 + 8\alpha^3 + \frac{4}{9}\alpha^4$$

$$- \frac{16}{3}\alpha^5 + \frac{16}{9}\alpha^6$$

$$|\lambda_4|^2 = 1 - \frac{16}{3}\alpha^2 + \frac{76}{9}\alpha^4 - \frac{32}{9}\alpha^6 + \frac{4}{9}\alpha^8$$

and

$$L = 4\Delta x$$

$$|\lambda_1|^2 = 1 - 2\alpha + 2\alpha^2$$

$$|\lambda_2|^2 = 1 - \alpha^2 + \alpha^4$$

$$|\lambda_3|^2 = 1 - \frac{2}{3}\alpha - \frac{1}{9}\alpha^2 + \frac{4}{3}\alpha^3 - \frac{1}{9}\alpha^4 - \frac{2}{3}\alpha^5 + \frac{2}{9}\alpha^6$$

$$|\lambda_4|^2 = 1 - \frac{5}{9}\alpha^2 + \frac{29}{36}\alpha^4 - \frac{5}{18}\alpha^6 + \frac{1}{36}\alpha^8.$$

For small α , these functional dependences demonstrate that the even-ordered schemes have similar dissipation for short scales, and that this dissipation is much less than that of the odd-ordered schemes (which also have similar dissipation). Figs. 1–4 show this to be true for the entire decade of wavelengths just above the Nyquist cutoff.

4. Comparison to other dissipative schemes

Many Eulerian differencing schemes which have been used for the advection problem are also nonconservative, and it is of interest to compare their dissipation rates to that of the semi-Lagrangian scheme. Three representative schemes are upstream differencing, the Lax–Wendroff scheme, and the Matsuno (Euler-backward) scheme (see Haltiner and Williams 1980, or O'Brien 1986, for examples and derivations). It is easy to see from its derivation that upstream differencing is exactly the first-order semi-Lagrangian technique (restricted to being nonmultiply-upstream). This technique produces an equivalent Laplacian (k^2) friction (the same as (3.3)) which is generally considered too dissipative for large-scale atmospheric or oceanic models.

a. Lax-Wendroff

It can be shown that the Lax-Wendroff scheme in its usual form is simply a second-order, nonmultiply upstream semi-Lagrangian technique. The k^4 spectrum is somewhat surprising, as the Lax-Wendroff scheme can be written in a form that appears to contain Laplacian (k^2) friction with a coefficient of $\frac{1}{2}U^2\Delta t$. The advection equation

$$\psi_t = -U\psi_x,$$

defines $\frac{\partial}{\partial t}$, so that we may also write

$$\psi_{tt} = U^2\psi_{xx}.$$

The Taylor series expansion for the temporal evolution of ψ is

$$\psi^{n+1} = \psi^n + \Delta t\psi_t + \frac{\Delta t^2}{2}\psi_{tt} + O(\Delta t^3).$$

Substituting in the definitions for ψ_t and ψ_{tt} gives

$$\psi^{n+1} = \psi^n - \Delta t U\psi_x + \frac{\Delta t^2}{2}U^2\psi_{xx} + O(\Delta t^3).$$

Although this equation appears to be a first-order (in time) advection coupled with a Laplacian dissipation, it is better understood as a quadratic interpolation formula for the value of ψ at $x - U\Delta t$, and is actually second-order in time, with a dissipation that depends on the accuracy of the spatial derivatives. The usual Lax-Wendroff scheme uses centered second-order spatial discretizations for ψ_x and ψ_{xx} , resulting in exactly the same transfer function as for the quadratic semi-Lagrangian case. Higher-order spatial discretizations yield cubic and quartic semi-Lagrangian schemes. Increasing the number of terms in the original Taylor series in time increases the temporal accuracy beyond $O(\Delta t^2)$. It should be noted here that in multiple spatial dimensions, the straightforward Lax-Wendroff scheme will differ from the semi-Lagrangian scheme using Lagrangian interpolation because Lagrangian interpolation introduces some extraneous terms to the Taylor series expansion to allow separability.

b. Matsuno (Euler-backward)

The Matsuno scheme is a well-known two-step integration technique, which can be written

$$\psi_j^* = \psi_j^n - \frac{U\Delta t}{2\Delta x}(\psi_{j+1}^n - \psi_{j-1}^n)$$

$$\psi_j^{n+1} = \psi_j^n - \frac{U\Delta t}{2\Delta x}(\psi_{j+1}^* - \psi_{j-1}^*)$$

The amplification factor is

$$|\lambda|^2 = 1 - (\alpha \sin k\Delta x)^2 + (\alpha \sin k\Delta x)^4.$$

For long waves, this reduces to

$$|\lambda|^2 = 1 - \alpha^2(k\Delta x)^2 + O(k^4).$$

The equivalent Laplacian friction coefficient for these long waves is

$$A_H = \frac{1}{2\Delta t} \alpha^2 \Delta x^2.$$

For long waves the dissipation acts like Laplacian friction, while for shorter waves the higher-order terms act to decrease the order of the dissipation, ultimately bringing the scheme back to neutral stability at $k\Delta x = \pi$. This Laplacian friction has a form very similar to that of the linear semi-Lagrangian technique (or upstream differencing), except that it depends on α^2 , rather than α . For a Courant number of 1, the friction reaches its upper bound of

$$\frac{1}{2} \frac{\Delta x^2}{\Delta t} = 5.6 \times 10^6 \text{ m}^2 \text{ s}^{-1}.$$

5. Discussion

a. Dependence on Δt

The expressions for the amplification factors in sections 2 and 3 all depend on Δt (through the α term). For a constant advecting velocity, an expression can be derived for the ratio of the decay time scales for various choices of the time step. For convenience, the time step will be halved relative to a standard case, and the decay time scales for the case of quadratic interpolation will be compared.

The ratio of the decay time scales is simply

$$\frac{\tau(\Delta t)}{\tau\left(\frac{\Delta t}{2}\right)} = \frac{\Delta t}{\ln\lambda(\Delta t)} \frac{\ln\lambda\left(\frac{\Delta t}{2}\right)}{\frac{\Delta t}{2}}.$$

For long waves (small $k\Delta x$), the Taylor series expansion for quadratic interpolation gives

$$\begin{aligned} \frac{\tau(\Delta t)}{\tau\left(\frac{\Delta t}{2}\right)} &= 2 \frac{\alpha_2^2(1 - \alpha_2^2)}{\alpha_1^2(1 - \alpha_1^2)} \\ &\quad - \frac{1}{4} \frac{\alpha_2^2(1 - \alpha_2^2)(\alpha_1^2 - \alpha_2^2)(1 - \alpha_1^2 - \alpha_2^2)}{\alpha_1^2(1 - \alpha_1^2)} \\ &\quad \times (k\Delta x)^4 + O(k\Delta x)^6, \quad (5.1) \end{aligned}$$

where α_1 and α_2 are defined below. Let $\hat{\alpha}$ be the Courant number; then, let α_1 denote the residual Courant number for the standard case, and α_2 denote the residual Courant number for the half-time-step case. These latter two quantities are defined by

$$\begin{aligned} \alpha_1 &= \hat{\alpha} - \text{NINT}(\hat{\alpha}) \\ \alpha_2 &= \hat{\alpha}/2 - \text{NINT}(\hat{\alpha}/2), \quad (5.2) \end{aligned}$$

where $\text{NINT}(\quad)$ is the nearest integer function.

Given these definitions, the coefficient of the second term of (5.1) is small in all cases (typically two–three orders of magnitude smaller than the first term). Coupled with the requirement that $(k\Delta x) \ll 1$ for the expansion to be valid, it is clear that the first term governs the ratio. Figure 5 shows the base 10 log of the first term of (5.1) for a Courant number varying between 0 and 2. Negative values imply that the *shorter* time step is less dissipative, while positive values imply that the *longer* time step is less dissipative. The ratio goes to infinity at a Courant number of 1.0, since the time scale for decay of the long time step is infinite at that value.

In practice, it is typically observed that semi-Lagrangian advection schemes are less dissipative with longer time steps. In light of Fig. 5, this can be seen as implying that the majority of the flow is contained in the range $0.5 < \alpha < 1.5$, which is a reasonable range of values for the requirements of accuracy and efficiency. It is, of course, also possible to estimate a distribution of the Courant numbers and integrate it against the first term in (5.1) [also using (5.2)] to obtain a single value for the ratio of the decay time scales. Unless the distribution were strongly weighted to small Courant numbers, the results can be expected to show a decrease in damping with larger time steps.

b. Dependence on Δx

It has already been mentioned that the dissipative decay time scale is related to the grid spacing in such a way that the decay time scale is independent of Δx for a wavelength of a fixed number of grid intervals. For a fixed *absolute* wavelength, however, decreasing the grid spacing by a factor of p increases the decay time scales (i.e., decreases the damping) of the four

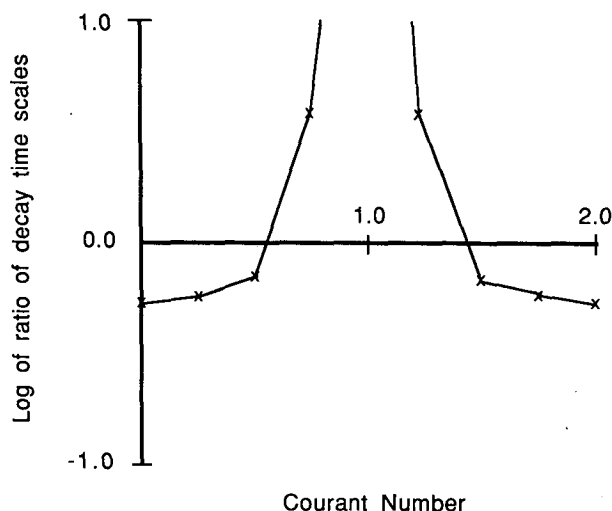


FIG. 5. Base 10 log of the ratio of the decay time scales when the time step is halved, as a function of the Courant number based on the longer time step.

schemes by factors of p^2 , p^4 , p^4 , and p^6 . It also doubles the Courant number, which may act to modify this improvement. This strong dependence of the effective viscosity on the grid spacing suggests that some care be taken with the choice of discretization parameters.

c. Shallow-water equation results

Ritchie (1988) has performed a series of experiments with a spectral shallow-water model which provide an excellent illustration of the applicability of the effective eddy viscosity concept in a more complex situation. He ran several spectral model simulations with various choices for the advection and diffusion. The Eulerian model was run with a biharmonic diffusion coefficient of $10^{15} \text{ m}^4 \text{ s}^{-1}$, and the semi-Lagrangian scheme used Lagrangian bicubic interpolation. Ritchie's Figs. 10 and 11 show that after 20 days, the T63 resolution semi-Lagrangian code lost about 5% of its total energy, which was about twice the dissipation of the Eulerian case. Increasing the resolution to T126 caused the energy losses in the semi-Lagrangian case to be negligible (<1%) over the 20-day integration period.

The T63 resolution case corresponds to a Δx of roughly 200 km, and was run with a time step of 3600 s. By this analysis, it should then display an equivalent biharmonic viscosity with a magnitude bounded above by $10^{16} \text{ m}^4 \text{ s}^{-1}$. The experiments suggest an effective viscosity of about $2 \times 10^{15} \text{ m}^4 \text{ s}^{-1}$, a reasonable agreement. The T126 experiments should have $1/16$ the viscosity ($1 \times 10^{14} \text{ m}^4 \text{ s}^{-1}$), and the observed viscosity was, in fact, seen to be small compared to $10^{15} \text{ m}^4 \text{ s}^{-1}$.

d. Application to ocean models

Large-scale ocean circulation models are run in two distinct regimes. Noneddy-resolving simulations typically use grid intervals of 100–400 km and require Laplacian eddy viscosities of the order of $10^4 \text{ m}^2 \text{ s}^{-1}$ to account for subgrid-scale processes. These models are run for periods of decades to centuries, and are used in climate studies. Eddy-resolving simulations have grid spacings in the range of 20–40 km and are typically used with Laplacian eddy viscosities of the order of $10^2 \text{ m}^2 \text{ s}^{-1}$. These models are run for months to years, and are used in dynamical studies.

Semi-Lagrangian advection with linear interpolation requires extremely fine grids to produce numerical diffusion values comparable to those quoted above. Assuming that the time step will be chosen such that an advection of 1 m s^{-1} produces a Courant number of 1, (3.3) yields $A_H = 1/8 \Delta x$. For noneddy-resolving simulations, this specifies a grid interval of 80 km and a time step of 22 h, while for the eddy-resolving case, it specifies a grid interval of 0.8 km and a time step of 13 min. For both regimes the grid spacings are too small, particularly in the latter case.

Modeling subgrid-scale dissipation by biharmonic diffusivity is a largely ad hoc method. Therefore, the

values to be used for the coefficients are much more difficult to assign than for the Laplacian case. In the absence of clear knowledge of the physical dissipation, and noting that the wind-driven ocean circulation is a forced, almost inviscid system, it should at least be required that the decay time scales should be longer than the time scales for variability of the forcing. For the wind-driven flow, the dominant signals are seasonal, so the dissipative decay time scales should be greater than order 100 days for an accurate simulation. Assuming quadratic or cubic interpolation, Fig. 2 suggests that the grid spacing be less than $1/10$ (preferably $1/20$) of the shortest wavelength one desires to model accurately. This scaling is the same as is typically suggested for accuracy considerations; however, computational resources are not generally available to perform simulations with this level of resolution.

6. Summary

For the simple case of constant advecting velocity, the dissipation due to semi-Lagrangian advection has been quantified and compared to the dissipation of other nonconservative integration schemes. The use of linear interpolation results in dissipation which is effectively Laplacian, while quadratic and cubic interpolation both result in biharmonic dissipation. The coefficients of Laplacian and biharmonic viscosity are shown to be strongly dependent on the resolution, so that care must be taken for the use of this method in long-term integrations. The formulae presented here [particularly (3.3) and (3.4)] can be used to obtain a good estimate for the resolution required to attain the desired degree of conservation in a simulation. For a completely conservative integration, the noninterpolating method of Ritchie (1986) is available.

This paper does not address the errors in the amplification factors due to incorrect estimation of the departure point, but McDonald (1987) provides a detailed analysis of the order of the errors from that source. The variability in the dissipation due to time- and space-varying velocity fields has also not been addressed. Since this variability would be very difficult to predict or monitor, the semi-Lagrangian technique should not be used as the sole source of viscosity in a viscous flow simulation. It can be used as the sole source of dissipation in a simulation of (nearly) inviscid flow, in which the viscosity is only present to prevent non-

linear instability (e.g., Ritchie 1988; Côté and Staniforth 1988).

Figures 1–4 show that the decay time scale as a function of the residual Courant number, α , is near its shortest (i.e., most dissipative) value over most of the range of α , so that the worst-case viscosity coefficients calculated in section 3a may be typical values if a broad range of residual Courant numbers are present.

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